B. Figures

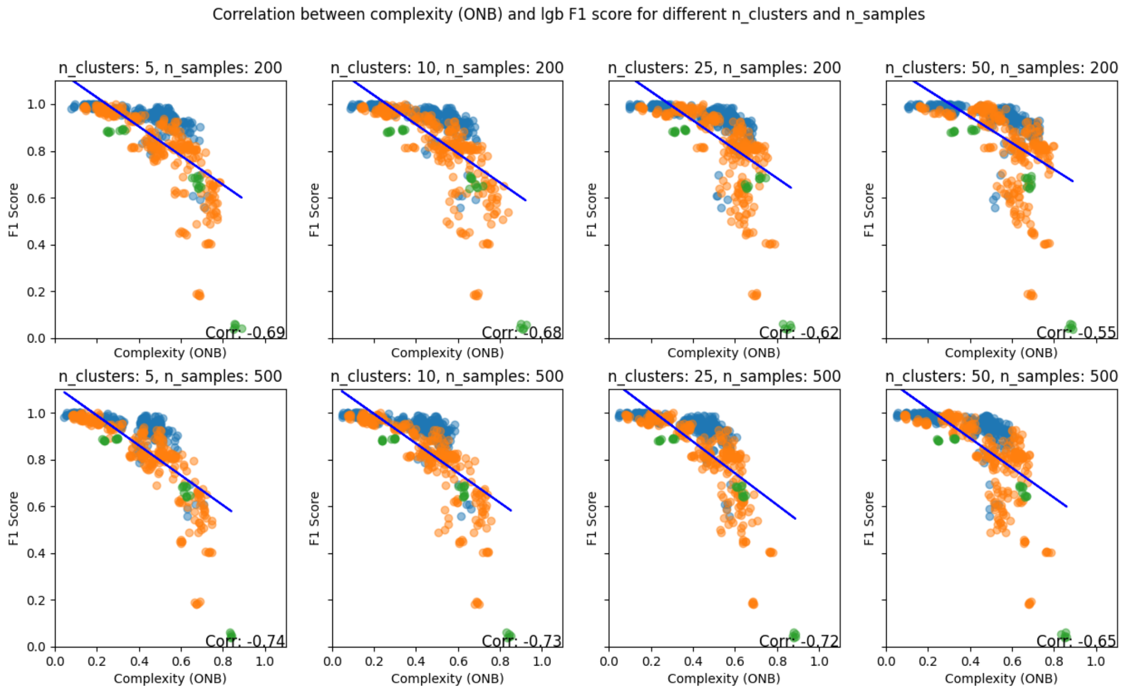

Fig. 1: Overlap (ONB) versus performance (in multiclass datasets) of LGB classifier for all pairs of clusters/samples. Each dot represents a dataset, with dots of the same color being different derived versions of the same dataset. The correlation value is displayed at the bottom.

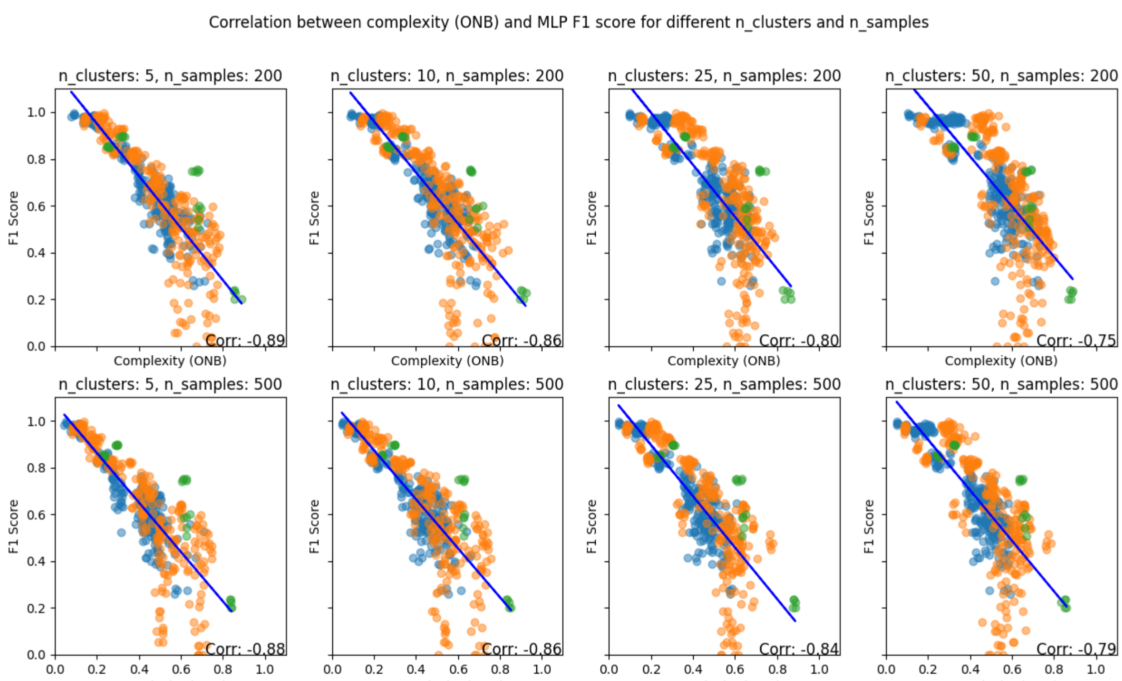

Fig. 2: Overlap (ONB) versus performance (in multiclass datasets) of MLP classifier for all pairs of clusters/samples. Each dot represents a dataset, with dots of the same color being different derived versions of the same dataset. The correlation value is displayed at the bottom.

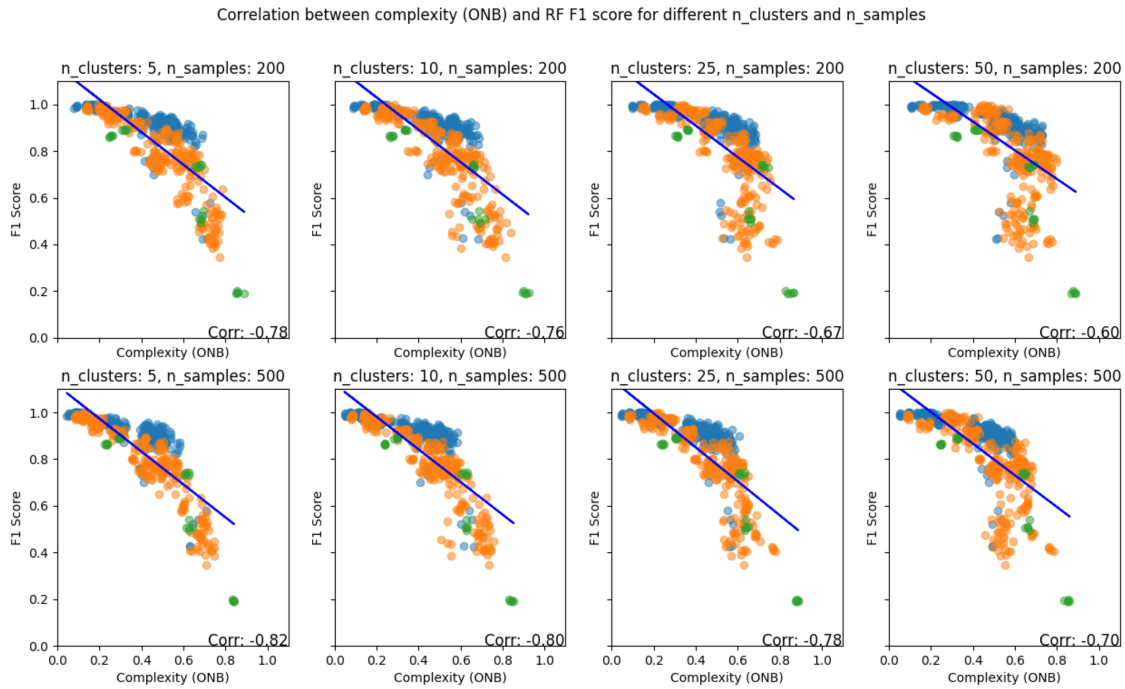

Fig. 3: Overlap (ONB) versus performance (in multiclass datasets) of RF classifier for all pairs of clusters/samples. Each dot represents a dataset, with dots of the same color being different derived versions of the same dataset. The correlation value is displayed at the bottom.

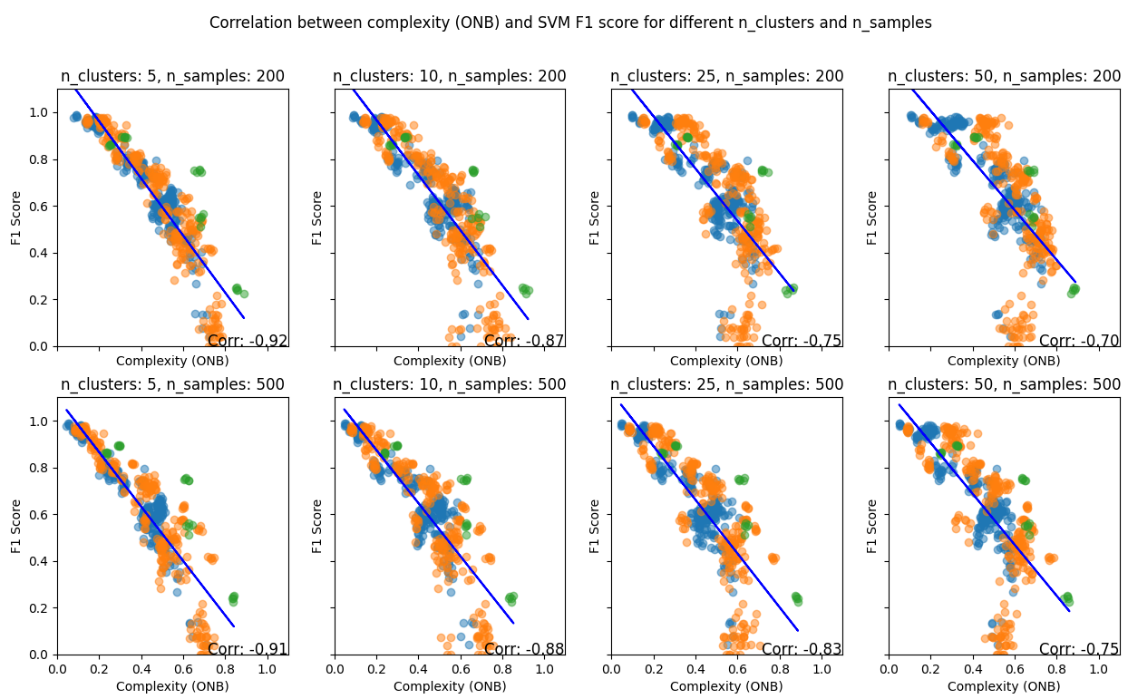

Fig. 4: Overlap (ONB) versus performance (in multiclass datasets) of SVM classifier for all pairs of clusters/samples. Each dot represents a dataset, with dots of the same color being different derived versions of the same dataset. The correlation value is displayed at the bottom.

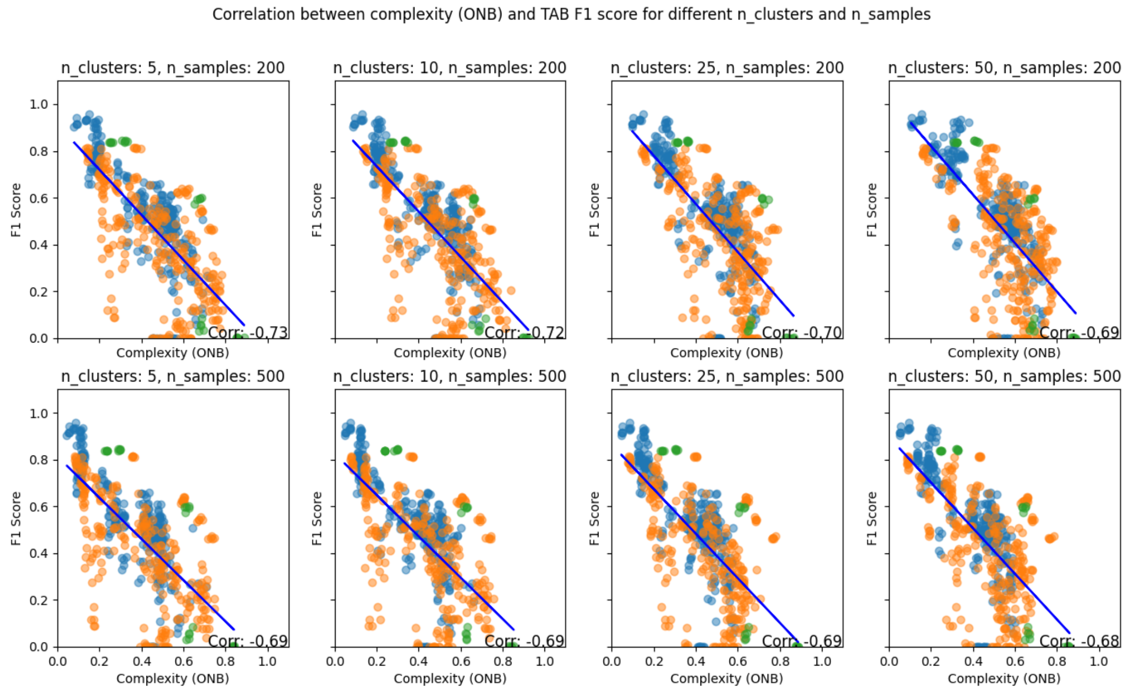

Fig. 5: Overlap (ONB) versus performance (in multiclass datasets) of TAB classifier for all pairs of clusters/samples. Each dot represents a dataset, with dots of the same color being different derived versions of the same dataset. The correlation value is displayed at the bottom.

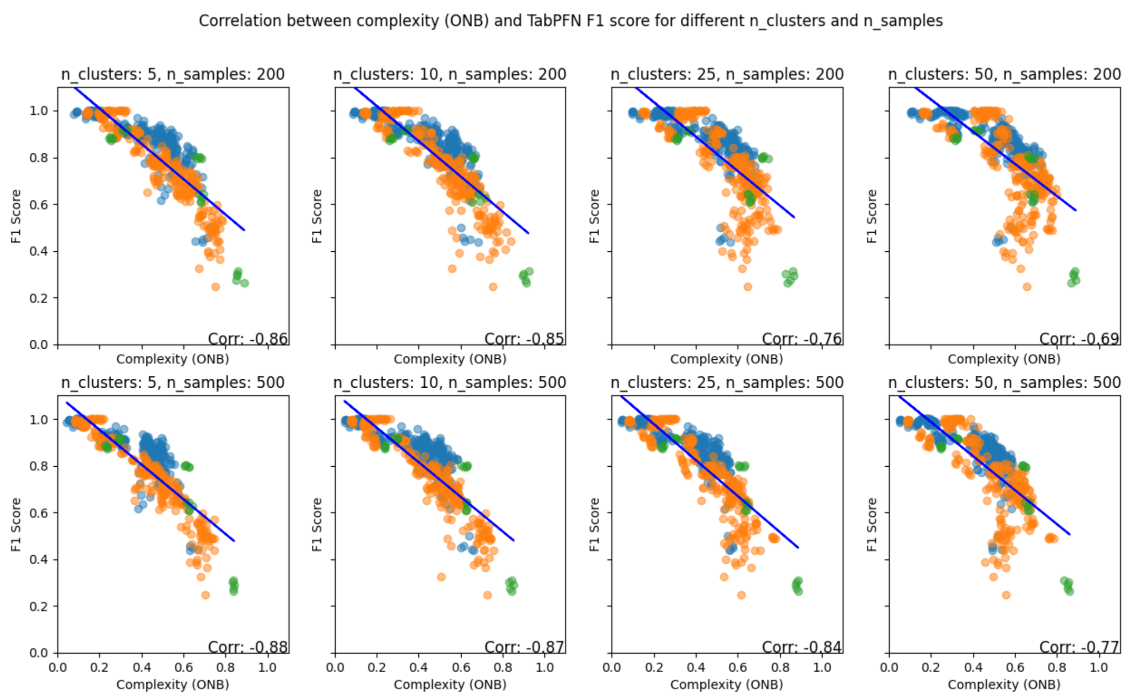

Fig. 6: Overlap (ONB) versus performance (in multiclass datasets) of TabPFN classifier for all pairs of clusters/samples. Each dot represents a dataset, with dots of the same color being different derived versions of the same dataset. The correlation value is displayed at the bottom.

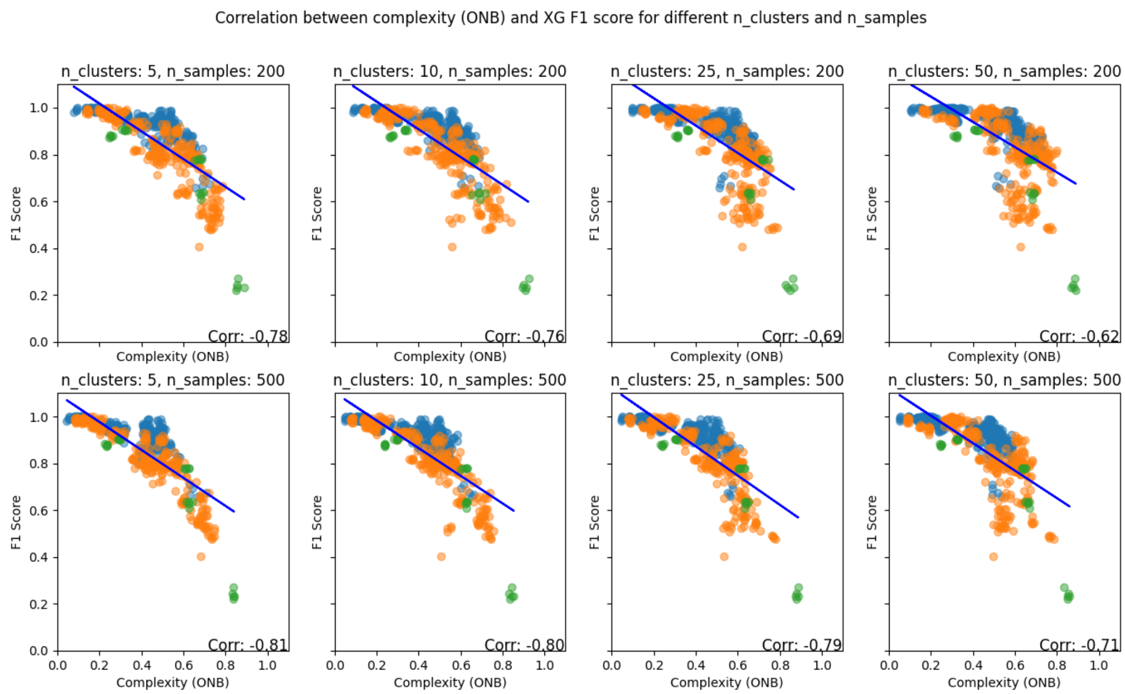

Fig. 7: Overlap (ONB) versus performance (in multiclass datasets) of XGB classifier for all pairs of clusters/samples. Each dot represents a dataset, with dots of the same color being different derived versions of the same dataset. The correlation value is displayed at the bottom.

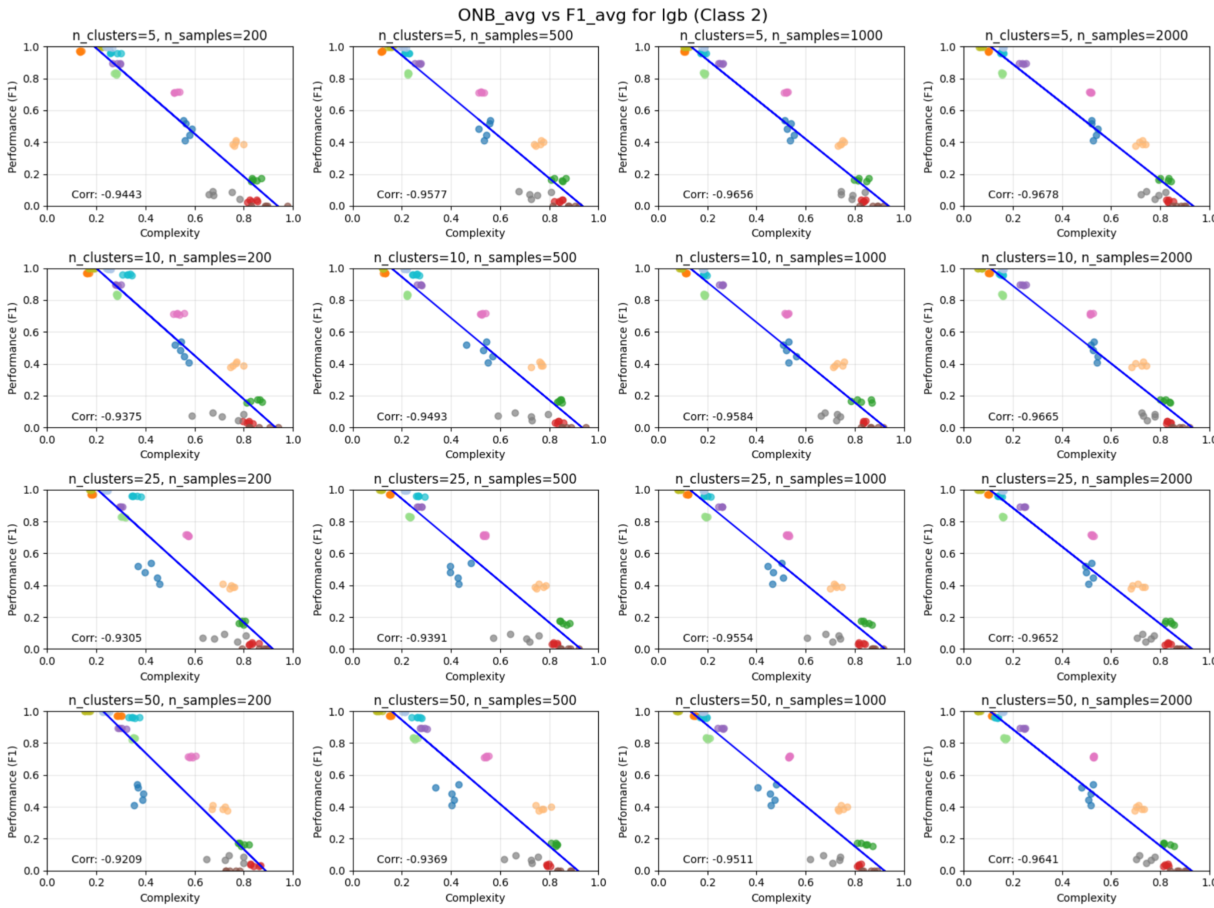

Fig. 8: Overlap (ONB) versus performance (in binary datasets) of lgb classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

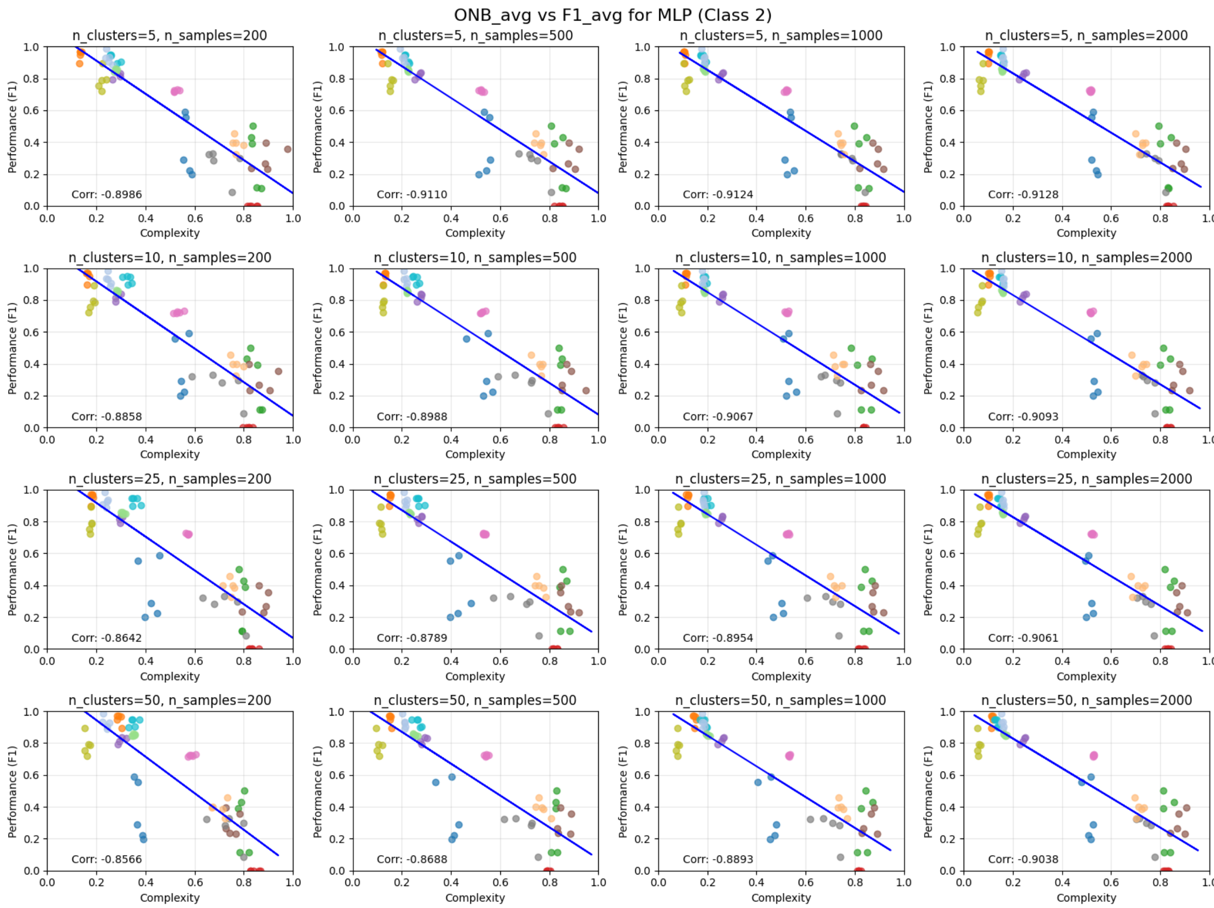

Fig. 9: Overlap (ONB) versus performance (in binary datasets) of MLP classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

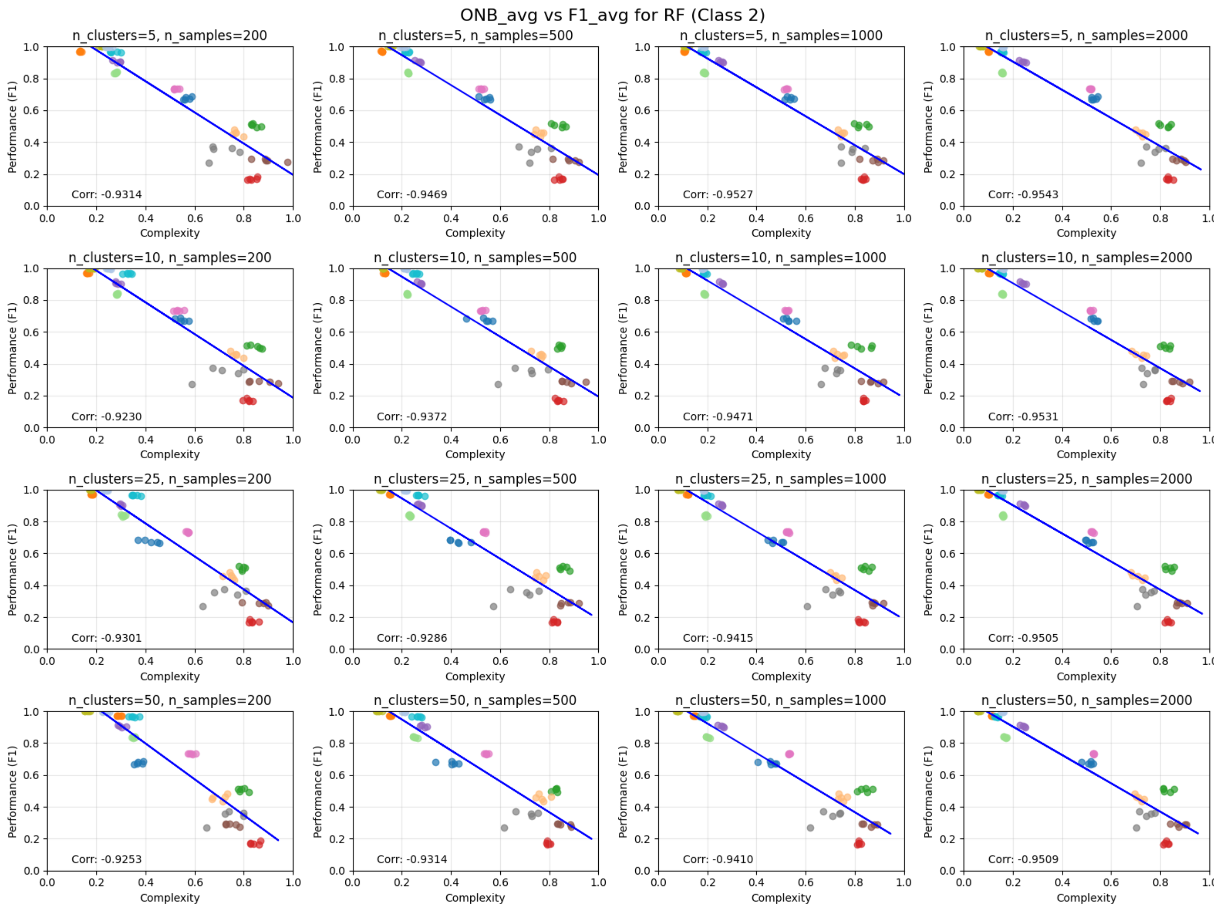

Fig. 10: Overlap (ONB) versus performance (in binary datasets) of RF classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

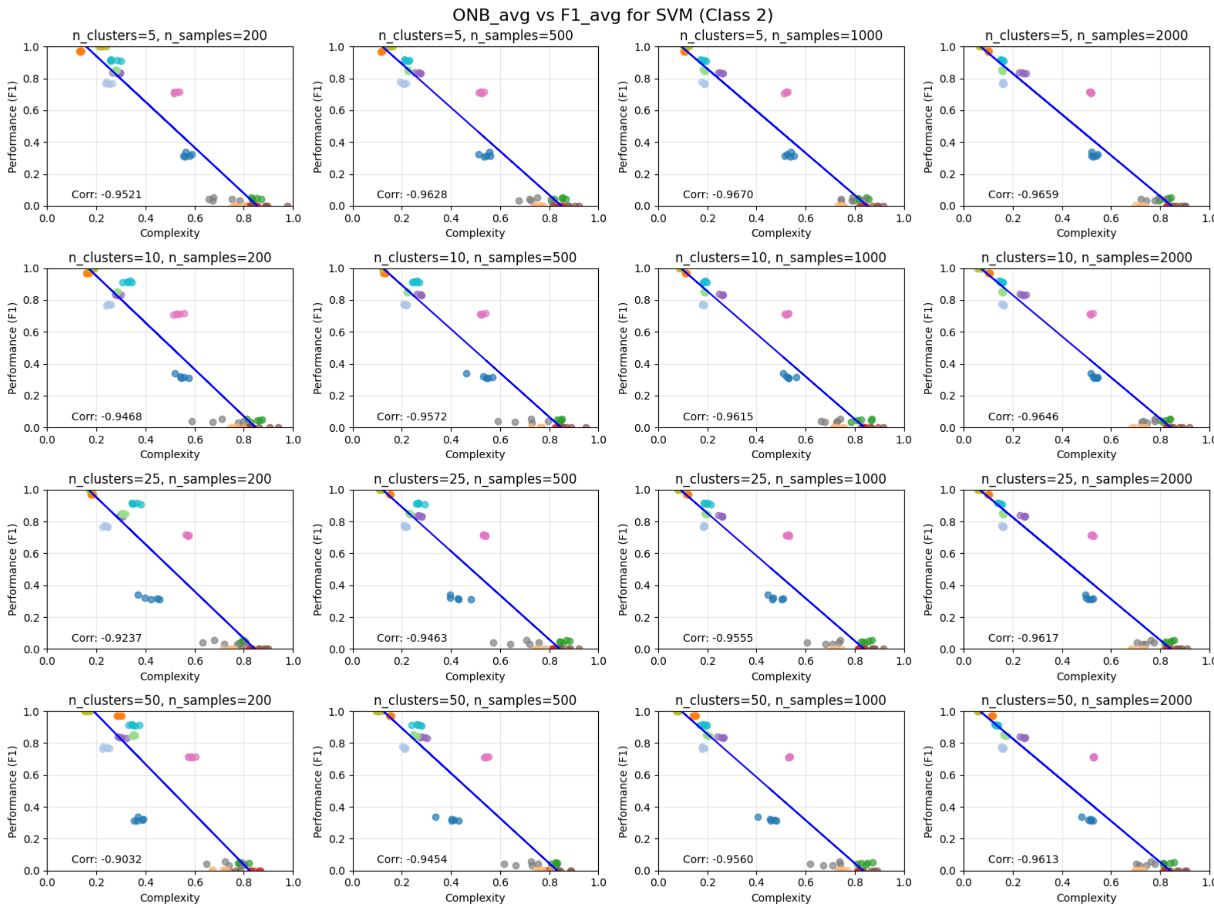

Fig. 11: Overlap (ONB) versus performance (in binary datasets) of SVM classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

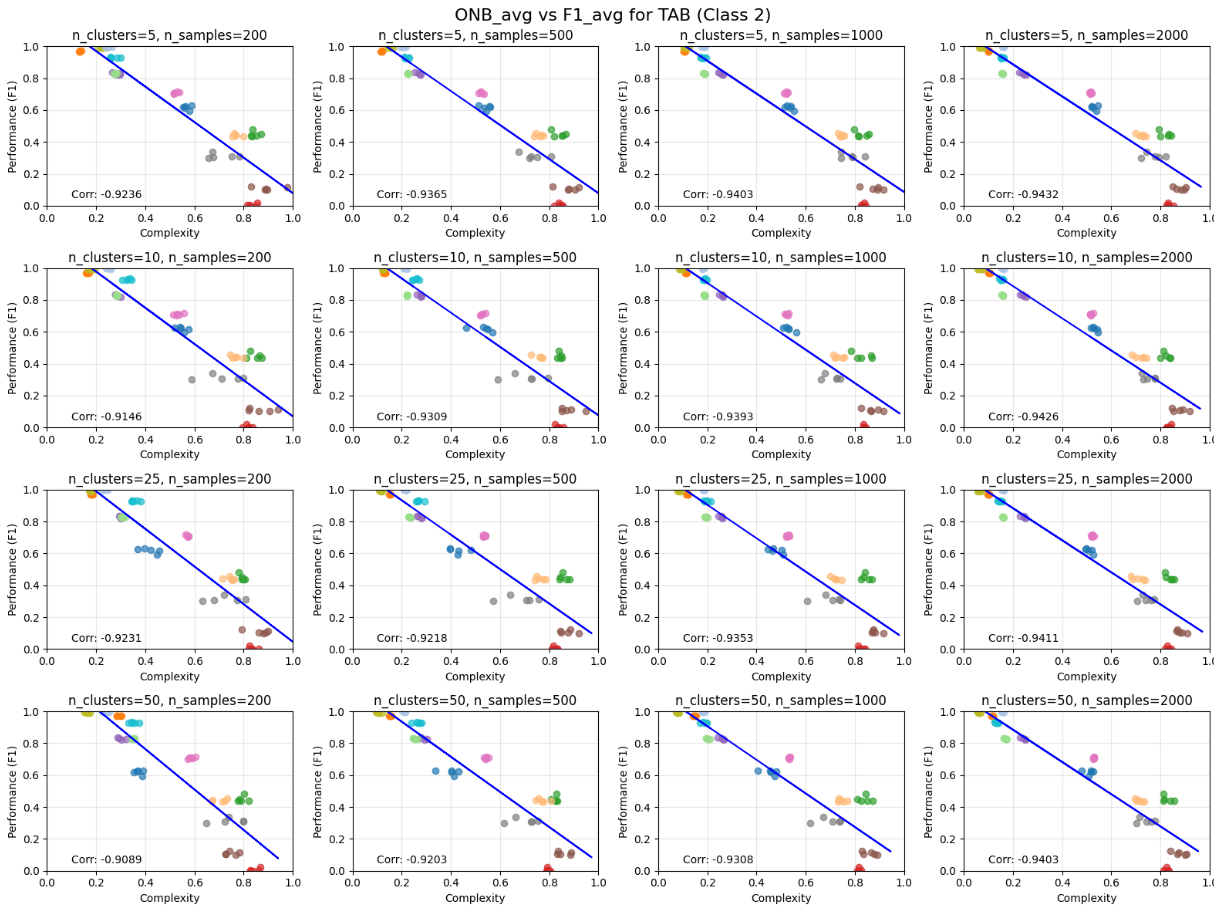

Fig. 12: Overlap (ONB) versus performance (in binary datasets) of TabNet classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

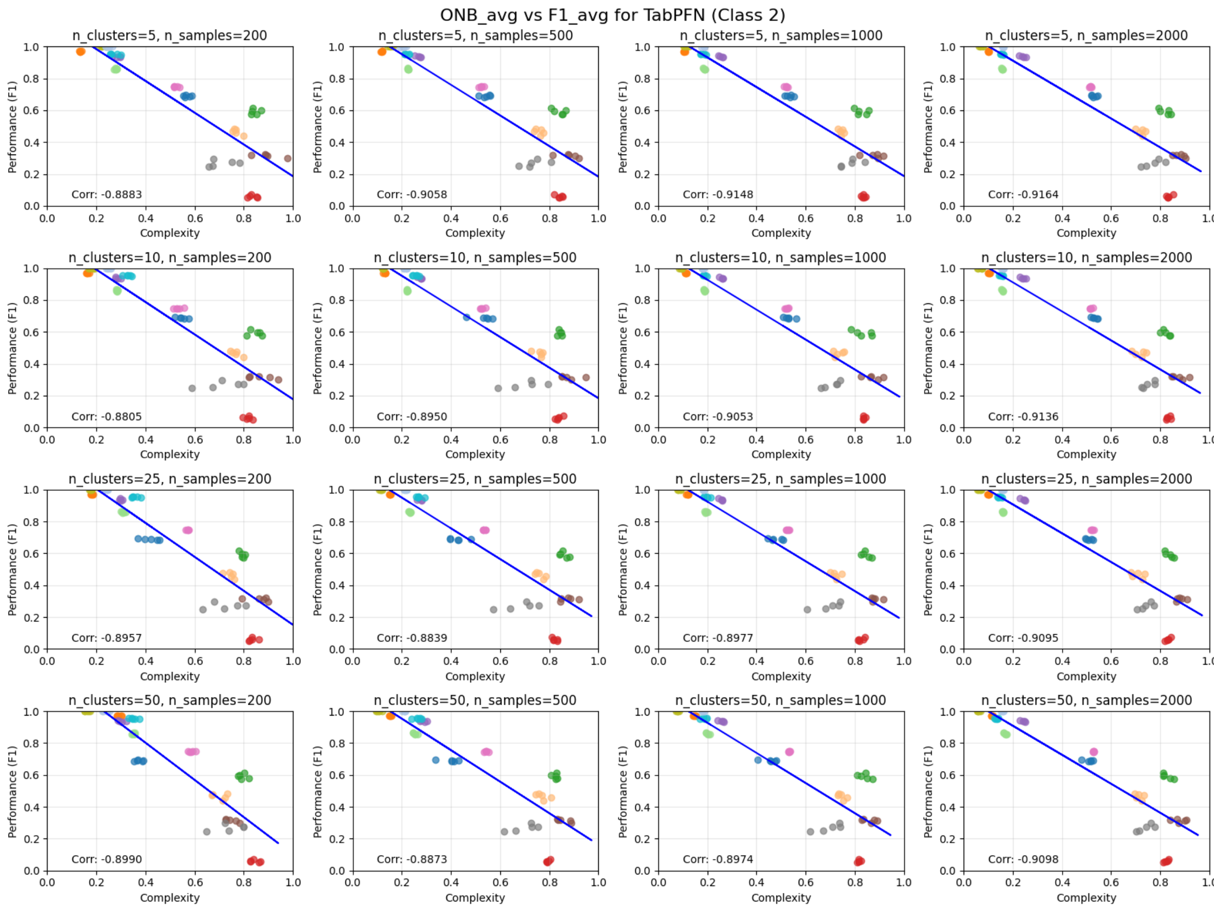

Fig. 13: Overlap (ONB) versus performance (in binary datasets) of TabPFN classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

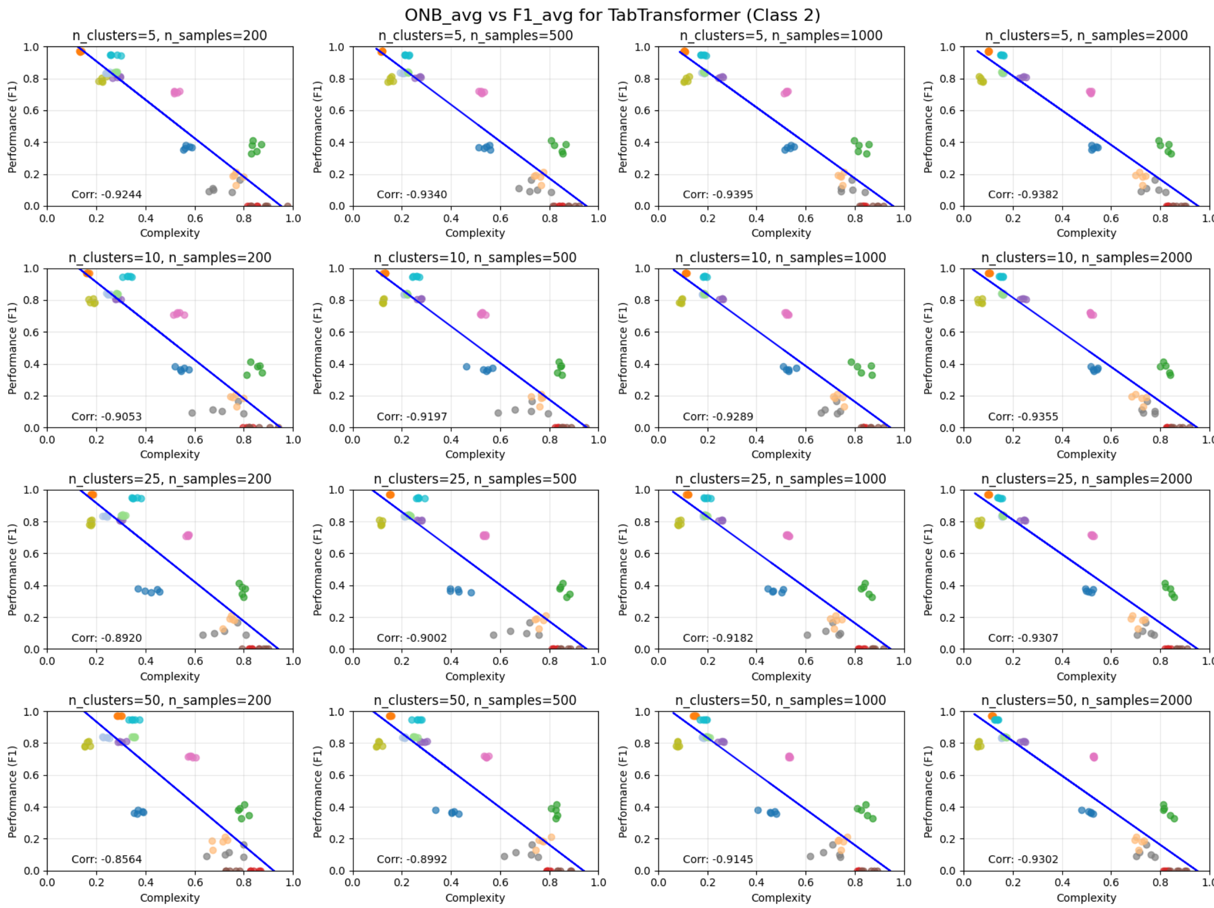

Fig. 14: Overlap (ONB) versus performance (in binary datasets) of TabTransformer classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

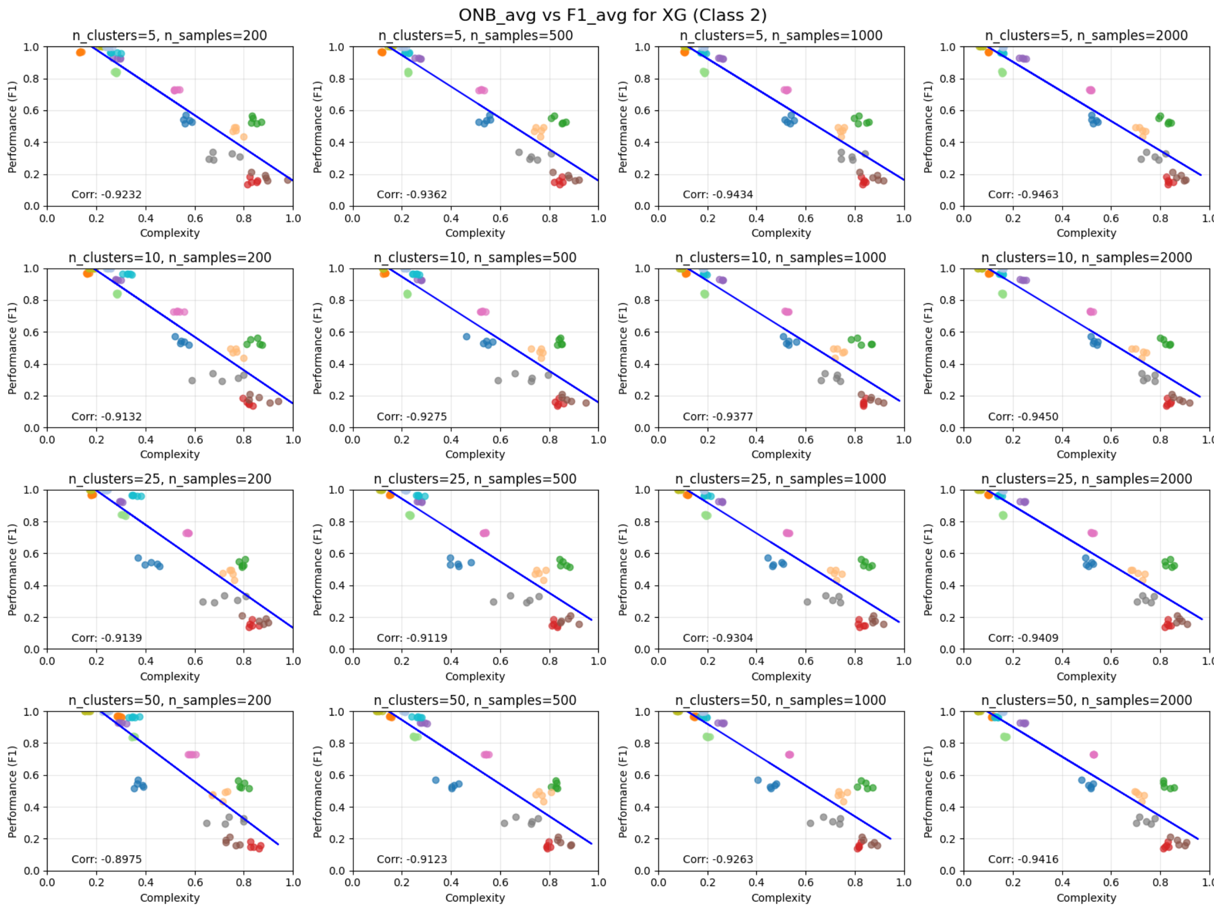

Fig. 15: Overlap (ONB) versus performance (in binary datasets) of XGB classifier for all combinations of number of clusters and number of samples. Each dot represents a dataset, with dots of the same color being different partitions of the same dataset obtained through cross validation. The plots demonstrate the linear relationship between the classification performance and estimated complexity. Specifically, the higher complexity, the lower F1 classification performance.

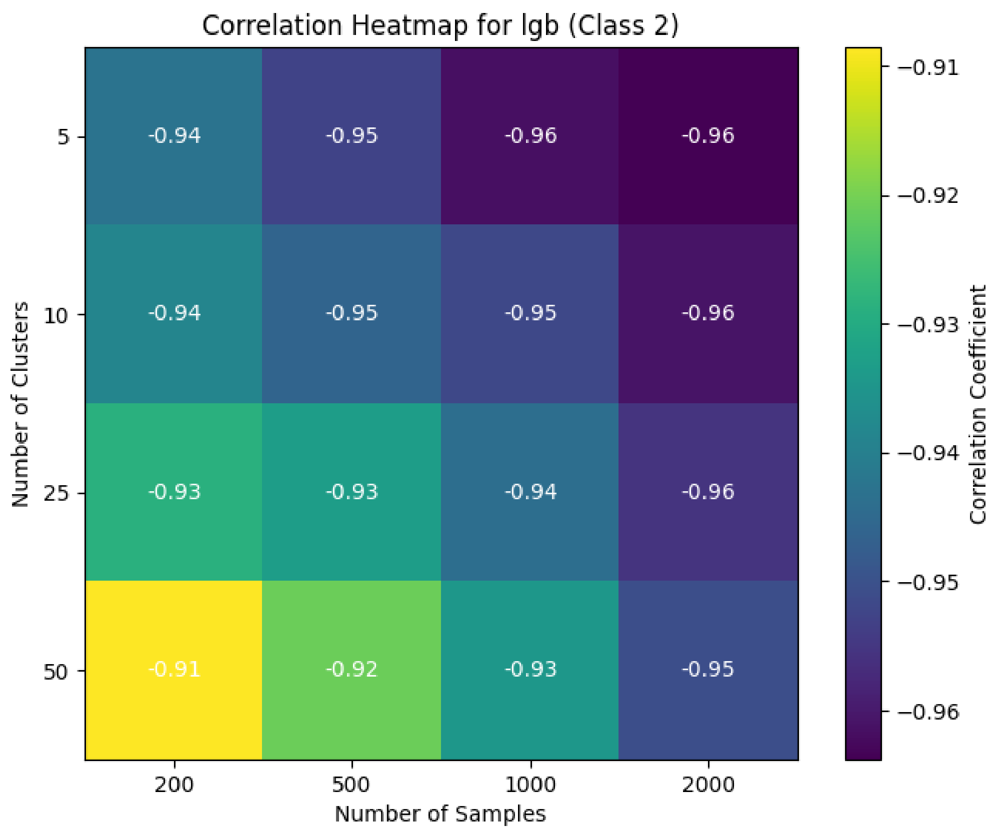

Fig. 16: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (lgb)

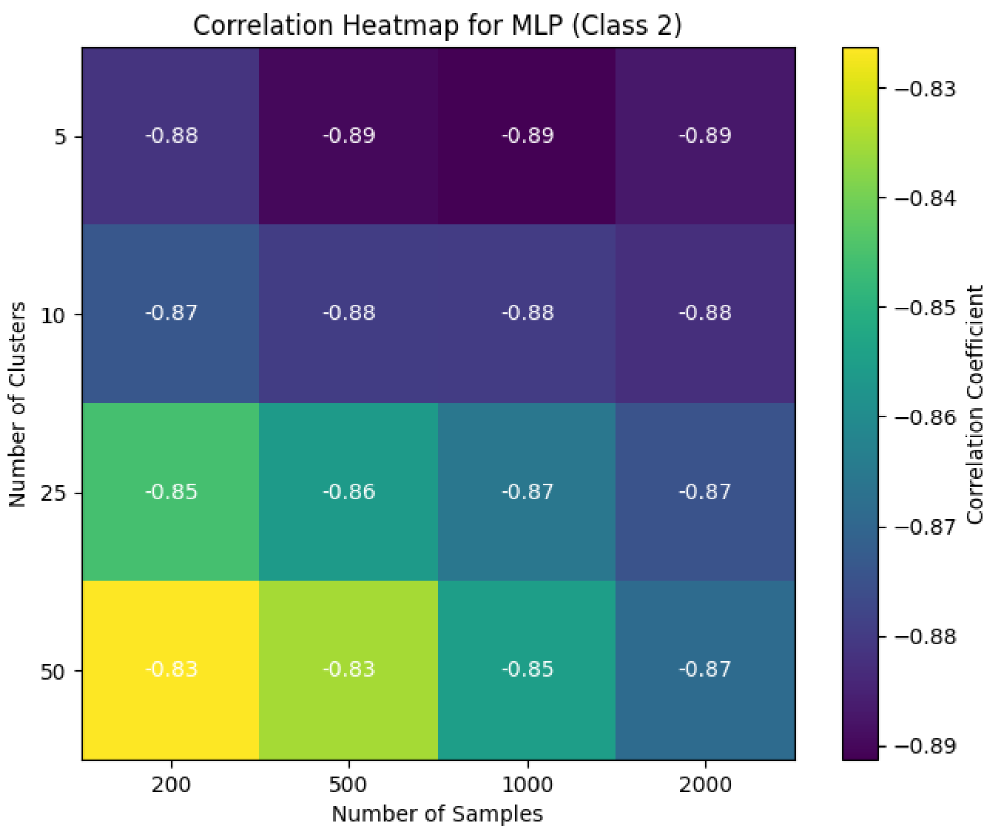

Fig. 17: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (MLP)

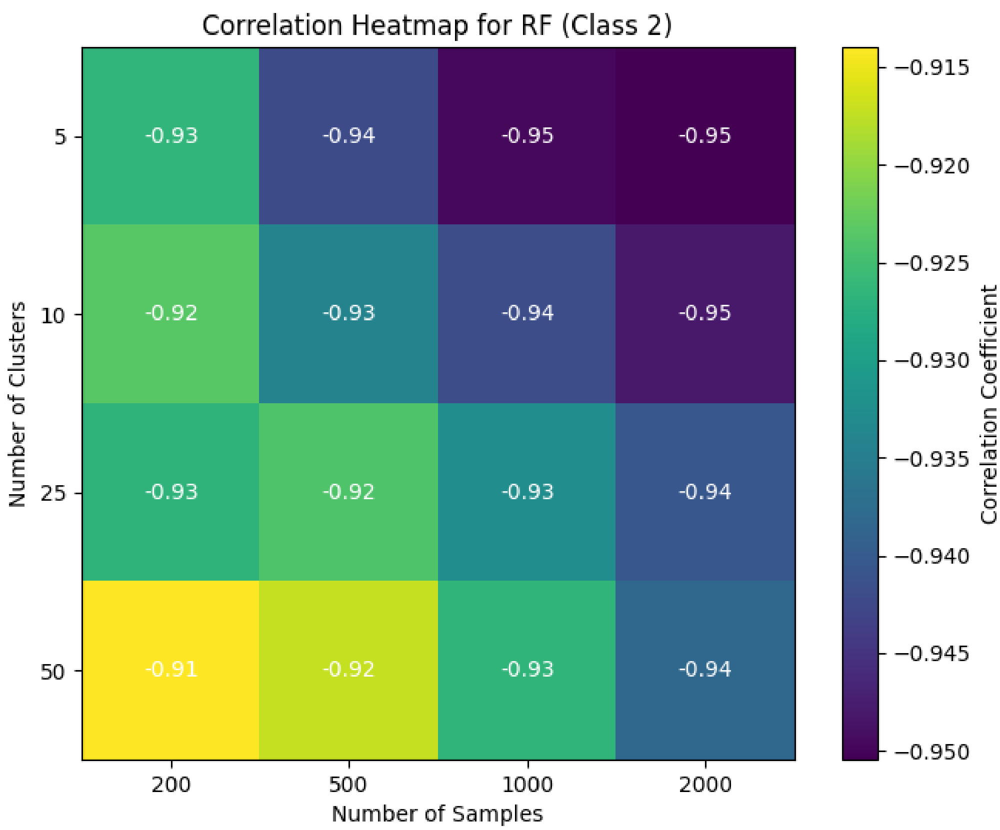

Fig. 18: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (RF)

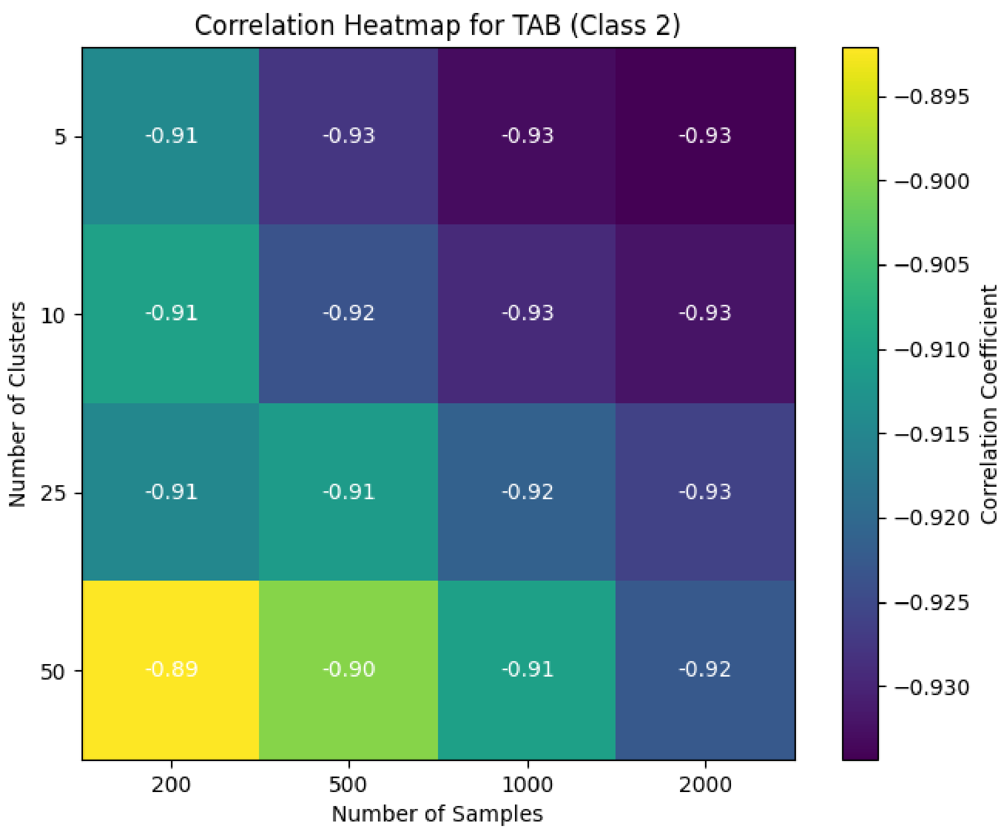

Fig. 19: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (TAB)

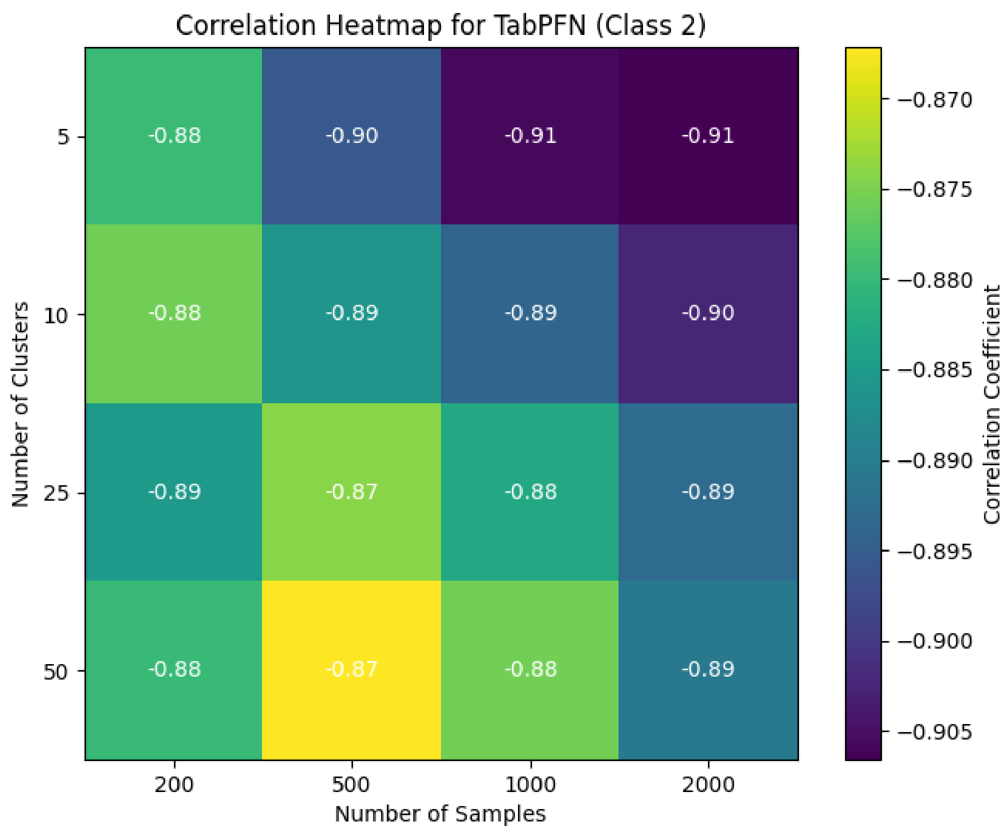

Fig. 20: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (TabPFN)

Fig. 21: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (TabTransformer)

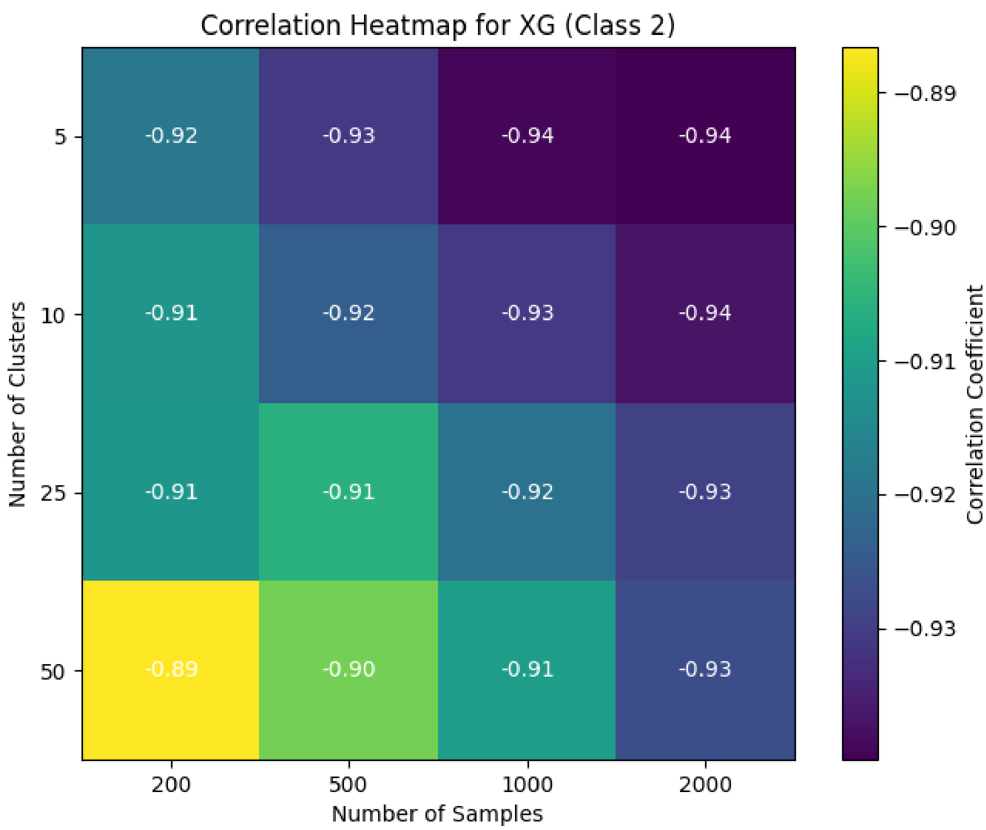

Fig. 22: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (XGB)

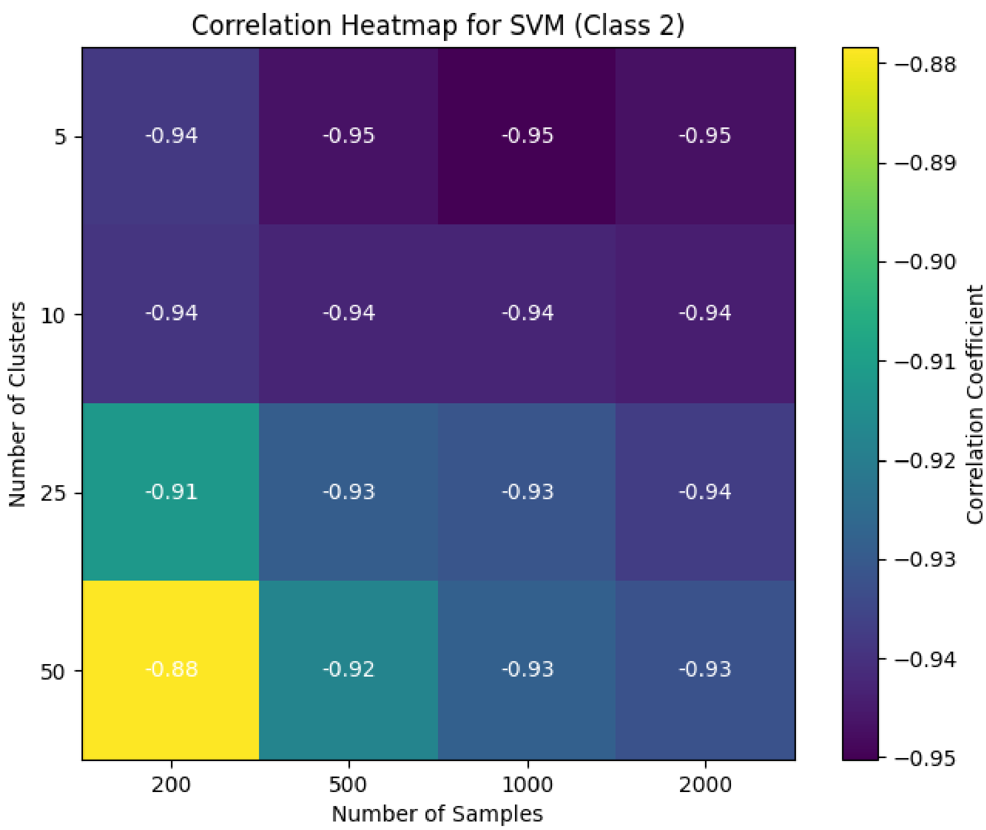

Fig. 23: Heatmap of correlation between Classification Performance (F1-Score) and Overlap (kDN) for each samples/cluster pair tested (SVM)